

UNIVERSITY OF ENGINEERING & TECHNOLOGY, TAXILA



Term Project

Parametric Analysis of Milling using RSM Central Composite Design
(CCD) method

Submitted To

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Submitted By

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Course

METAL FORMING & CUTTING ANALYSIS (IE-416)

INDUSTRIAL ENGINEERING DEPARTMENT

Complex Engineering Problem

Problem Statement:

It is desired to perform the “Parametric Analysis of Milling using RSM Central Composite Design (CCD) method”.

Background:

Parametric analysis in milling refers to the systematic study and optimization of various machining parameters (such as spindle speed, feed rate, depth of cut, and cutting tool geometry) to achieve optimal performance in terms of productivity, quality, and cost. The goal is to understand the relationship between input parameters and the output responses (such as surface roughness, material removal rate and machining time) to improve the efficiency of the milling process.

One effective technique for performing such analysis is **Response Surface Methodology (RSM)**, particularly the **Central Composite Design (CCD)** using MINITAB Statistical Software, which is widely used for process optimization in the context of milling operations.

Parametric analysis of milling using RSM and CCD provides a powerful framework for optimizing milling operations by systematically analyzing how different machining parameters influence the desired outcomes. By leveraging these techniques, engineers and operators can not only improve efficiency and quality but also gain valuable insights into the interactions between various process variables. The result is an optimized and robust milling process that enhances both productivity and product quality.

Objectives:

The primary objective of this study is to perform **parametric analysis of the milling process** using **Response Surface Methodology (RSM)**, particularly the **Central Composite Design (CCD)**, to optimize the milling parameters (such as spindle speed, feed rate, and depth of cut) in order to enhance **material removal rate (MRR)**, minimize **surface roughness**, and optimize **machine time**.

Task to be performed:

- **literature Review and Experiment Design:** Conduct a literature review on parametric optimization in milling using RSM and CCD, then define the milling parameters (spindle speed, feed rate, depth of cut) and responses (surface roughness, MRR, machine time), followed by designing the experiment using Central Composite Design (CCD).
- **Data Collection and Analysis:** Perform the milling experiments based on the CCD matrix, then analyze the collected data using Response Surface Methodology (RSM) to fit a second-order regression model and interpret the effects of milling parameters on the responses.
- **Optimization and Validation:** Optimize the milling process to achieve the best trade-off

between MRR, surface roughness, and machine time, validate the optimized parameters through further experimentation, and prepare a final report with findings and recommendations.

Instructions:

- Student would work in group of 3-4 members.
- Students would select, and use required machine and measuring instruments in work shop.
- Students will complete analysis and submit project reports in word format.
- The report should reflect individuality and independent analysis of each group.
- The report submission and presentations will be done in 16th week of semester.

Assessment Sheet:

The assessment would be based on active participation, correct selection of machine parameters and measuring instruments, correctly recording of data, quality of analysis and timely submission of report.

Knowledge components	Domain	PLOs mapped	Taxonomy level	Contribution	Max. marks	Obtained marks
1. Each group member has actively participated in hands on experimentation.	Psychomotor	4	Manipulate (P2)	70%	6	
2. Each group has achieved required accuracy in performance and made size within the tolerance.		4	Precision (P3)		7	
3. Each group member worked collectively in form of team.	Affective	9	Respond (A2)	20%	2.5	
4. Each group has submit the project as per given timeline and divide each activity as per schedule.		11			2.5	
5. Each group has compiled data and analyzed the results by using modern engineering tools.	Cognitive	2	Analyze (C4)	10%	2	
				Total	20	

Undertaking

We certify that the project work titled “Parametric Analysis of Milling Using Response Surface Methodology (RSM) and Central Composite Design (CCD) to Optimize Milling Parameters” is our own work. A small portion of the work presented in this project has been submitted in support of another award or qualification either at this institution or elsewhere. Where material and milling machine data have been used from other sources, they have been properly acknowledged and referred.

Abstract

The study presents a comprehensive parametric analysis of milling processes utilizing Response Surface Methodology (RSM) with a focus on the Central Composite Design (CCD) approach. RSM is a powerful statistical technique employed to model and optimize complex processes by evaluating the relationships between multiple input variables and the resulting output responses. In this research, various milling parameters such as cutting speed, feed rate, and depth of cut were systematically varied to assess their impact on performance metrics, particularly surface roughness.

Acknowledgement

We would like to express our sincere gratitude to all those who contributed to the successful completion of this research project on the "Parametric Analysis of Milling using RSM Central Composite Design (CCD) method." First and foremost, we extend our heartfelt thanks to our course teacher Dr Muhammad Jawad, whose guidance and expertise were invaluable throughout this study. Their insightful feedback and encouragement motivated us to delve deeper into the complexities of milling processes and statistical methodologies. We also wish to acknowledge the support of our research team members, whose collaborative spirit and dedication were crucial in conducting the experiments and analyzing the data. Their commitment to excellence and teamwork made this project not only possible but also enjoyable. Special thanks are due to the technical staff at our institution for their assistance with the milling equipment and experimental setup. Their expertise ensured that we adhered to the highest standards of precision during our experiments. Additionally, we appreciate the resources provided by our institution, including access to laboratories and software tools necessary for implementing Response Surface Methodology. This support greatly enhanced our research capabilities. This research would not have been possible without the contributions of all these individuals, and we are profoundly grateful for their support.

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Chapter 01

Introduction

Depth of Knowledge Required (CEP-Attribute 3)

The milling process is critical in manufacturing, impacting both product quality and production efficiency. This project aims to conduct a parametric analysis of milling operations using the Central Composite Design (CCD) method within Response Surface Methodology (RSM). By systematically varying milling parameters—such as cutting speed, feed rate, and depth of cut—this study seeks to identify optimal conditions that maximize tool life and improve surface finish.

1.1 Importance of Milling in Manufacturing

Milling is a fundamental machining process that involves removing material from a workpiece using a rotating cutter. This subtractive manufacturing technique is pivotal in modern industrial applications, allowing for the creation of complex shapes and precise components across various materials, including metals, plastics, and composites. The versatility of milling makes it suitable for numerous applications, from prototyping to mass production. The significance of milling in manufacturing cannot be overstated. It serves as the backbone of many industries, including aerospace, automotive, and electronics. In aerospace, for example, milling is essential for producing lightweight and intricate parts required for flight-engineered products such as airframe components and engine parts. The precision offered by CNC (Computer Numerical Control) milling ensures that these components meet stringent tolerances necessary for safety and performance in high-stakes environments.

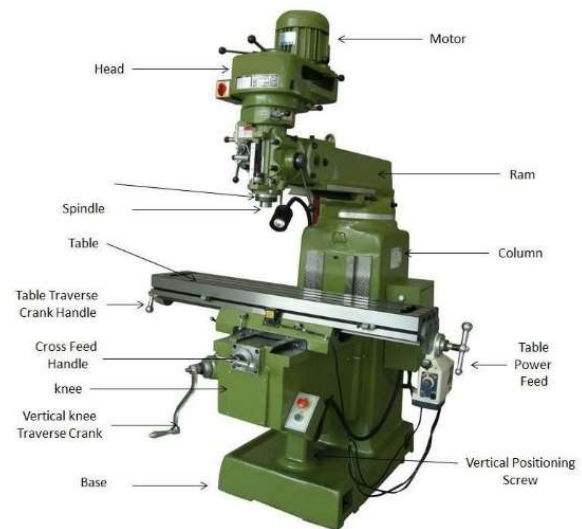


Figure 1: Schematic Diagram of Milling Machine

1.2 Efficiency and Cost-Effectiveness

Milling's efficiency is another significant advantage. The process allows for rapid production rates, making it ideal for both small-batch and high-volume manufacturing. CNC technology automates the milling process, reducing labor costs while increasing consistency and quality in part production. Moreover, the ability to produce parts with minimal waste enhances its cost-effectiveness compared to other manufacturing methods.

1.3 Evolution of Milling Technology

In addition to its operational benefits, milling has evolved significantly since its inception. The first milling machines were developed in the early 19th century to expedite the manufacturing of complex parts. Over the years, advancements in milling technology have led to the development of various types of milling operations—such as face milling, cylindrical milling, and finish milling—each tailored to specific applications and materials.

1.4 Objectives of the Study

This project focuses on utilizing Response Surface Methodology (RSM) with Central Composite Design (CCD) to optimize milling parameters effectively. By systematically analyzing factors such as cutting speed, feed rate, and depth of cut, this study aims to enhance tool life and surface finish quality. Understanding these parameters is crucial for improving overall machining performance and ensuring that manufacturers can meet the increasing demands for precision and efficiency in production processes.



Figure 2: Milling Experiment under observation

1.5 Relevance to Modern Manufacturing Trends

By leveraging RSM and CCD methodologies, this research not only contributes to the optimization of milling operations but also aligns with broader trends in manufacturing towards more efficient and effective production strategies. As industries continue to evolve with technological advancements, the insights gained from this study will be invaluable for practitioners seeking to refine their machining processes.

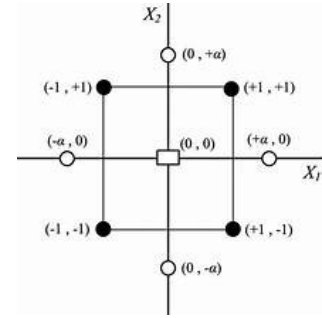


Figure 3: Schematic of RSM-CCD Plan

Chapter 02

Literature Review

Response Surface Methodology (RSM) is a powerful statistical tool used for optimizing processes by modeling and analyzing the relationships between several explanatory variables and one or more response variables. Among the various designs available within RSM, Central Composite Design (CCD) is particularly favored due to its efficiency in exploring the response surface with a limited number of experimental runs.

2.1 Applications of Central Composite Design in Machining

The application of CCD in machining optimization has been extensively documented in the literature. For instance, a study by **Thiagarajan et al. (2012)** utilized CCD to optimize drilling parameters for steel, demonstrating that this method effectively identifies optimal machining conditions while minimizing the number of experiments required. The research highlighted that using CCD, the maximum tool life achieved was **96.7 minutes**, surpassing the predicted value by **9.36%** based on a quadratic regression model developed from the experimental data. This study underscores CCD's capability to efficiently model complex interactions between cutting parameters such as cutting speed, feed rate, and depth of cut in milling operations. Similarly, Kasina et al. (2020) explored CCD's versatility across various applications, including optimizing

machine parameters in drilling hybrid metal matrix composites. Their findings indicated that CCD could effectively predict responses with high accuracy, reinforcing its applicability in different manufacturing contexts. The simplicity and efficiency of CCD make it a preferred choice among researchers seeking to estimate second-order models without requiring extensive experimental data.

2.2 Advantages of CCD Over Other Experimental Designs

The advantages of using CCD over other experimental designs, such as full factorial designs, are well-documented. Full factorial designs often require a substantial number of experimental runs to capture interactions among factors accurately, particularly when dealing with multiple variables. In contrast, CCD can achieve similar modeling success with fewer runs—typically around 20 experiments for three factors compared to 27 for full factorial designs. This efficiency is crucial in industrial settings where time and resources are limited.

Moreover, studies have shown that CCD provides high efficiencies under various optimality criteria (A-, D-, and E-optimality), making it a robust choice for process optimization. The ability to estimate all parameters in a full second-order model while maintaining simplicity is a significant advantage that has led to its widespread adoption in manufacturing research.

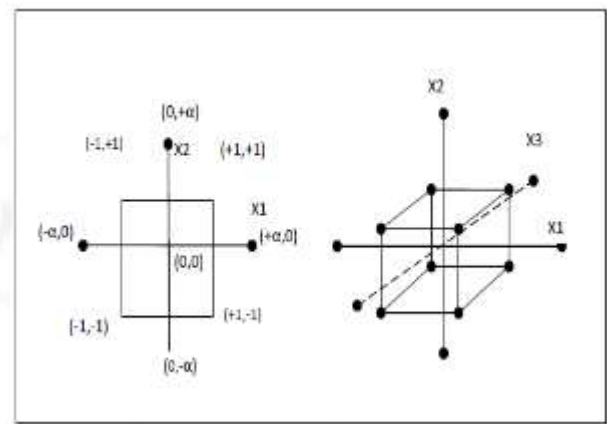


Figure 4: Central Composite Design in a box

2.3 Recent Developments and Future Directions

Recent advancements in RSM and CCD applications extend beyond traditional machining processes. For instance, research has applied these methodologies to optimize parameters in single-point incremental forming of titanium alloys, showcasing their relevance in modern manufacturing techniques that prioritize precision and material efficiency.

These studies emphasize the adaptability of CCD across different materials and processes, indicating its potential for future applications in emerging manufacturing technologies. In

conclusion, the literature demonstrates that Central Composite Design is an invaluable tool within Response Surface Methodology for optimizing machining processes. Its ability to efficiently model complex relationships with fewer experiments makes it particularly advantageous in industrial applications where resource optimization is critical. As manufacturing continues to evolve with new materials and technologies, further exploration of CCD's capabilities will likely yield significant insights into process improvements and innovations.

2.4 Integration of CCD with Other Optimization Techniques

Recent studies have explored the integration of **Central Composite Design (CCD)** with other advanced optimization techniques, such as **genetic algorithms (GA)** and **artificial neural networks (ANN)**, to improve the robustness and adaptability of the models. For example, a study by Khan et al. (2018) combined CCD with **genetic algorithms** to optimize cutting conditions in turning operations. This hybrid approach enabled more accurate predictions of tool life and surface finish compared to using CCD alone, especially when dealing with complex nonlinear behaviors in the machining process. The synergy between CCD and other methods enhances the capability of RSM in solving real-world machining problems with greater precision.

2.5 Application of CCD in Complex Machining Operations

CCD has proven effective in more complex machining operations, such as **multi-objective optimization** in **CNC machining** and **gear manufacturing**. In studies like those by Agarwal et al. (2016), CCD was used to optimize parameters in **multi-axis milling** and **gear cutting**, where multiple conflicting objectives (such as material removal rate and surface quality) had to be simultaneously optimized. These studies highlight the flexibility of CCD in handling intricate machining operations and its ability to model complex interactions between various factors, such as spindle speed, tool material, and cutting geometry.

2.6 Validation of CCD Models in Real-World Applications

One of the strengths of CCD is its ability to provide robust models that can be validated and implemented in real-world machining operations. For instance, in a study by Patel and Chauhan (2017), CCD was used to optimize milling conditions for **aluminum alloys**. The predicted values from the CCD model were compared with experimental results, showing a high degree of accuracy. This demonstrates that the models derived from CCD are not only theoretically sound

but can also be reliably used in actual production environments, ensuring the practical utility of RSM in industrial settings.

2.7 Role of CCD in Sustainability and Green Manufacturing

In the context of **sustainability**, CCD is increasingly being applied to optimize machining processes in ways that minimize energy consumption, tool wear, and material waste. Studies like those by Kumar et al. (2020) have focused on using CCD to optimize machining parameters for **dry machining** and **sustainable materials**, emphasizing the reduction of cutting fluid usage and minimizing environmental impact. As the demand for **green manufacturing** practices grows, CCD's role in helping manufacturers reduce their ecological footprint while maintaining optimal production efficiency is becoming more significant.

2.8 Practical Considerations and Challenges in CCD Application

Despite the many advantages of CCD, practical challenges remain in its application. One of the main issues involves the **selection of appropriate factors and levels** for experimental runs. In some cases, selecting too many factors or inappropriate levels can result in inefficient experiments or the inability to identify the most influential variables. Additionally, while CCD is highly effective for modeling second-order relationships, the **nonlinearity** of certain machining processes can still pose challenges. Research by Singh et al. (2019) highlights these challenges and suggests that combining CCD with alternative modeling approaches, such as **response surface methodology with machine learning (RSM-ML)**, could offer even more robust solutions.

2.9 Impact of CCD on Process Monitoring and Control

The application of CCD in **process monitoring and control** has gained attention in recent research. Studies have demonstrated that CCD can be used to design experiments that enhance the understanding of **real-time feedback systems** in CNC machines. By monitoring the relationships between cutting parameters and performance metrics, such as surface roughness and temperature, manufacturers can adjust machining conditions dynamically to improve part quality and reduce tool wear. CCD models, when integrated with **sensor-based monitoring systems**, have the potential to facilitate **autonomous machining** and predictive maintenance in Industry 4.0 environments.

2.10 CCD in Hybrid Machining and Multi-Process Optimization

The need for optimization in **hybrid machining processes**, such as **electrical discharge machining (EDM)** combined with milling, has been explored using CCD. These processes require careful coordination of various factors across different machining stages. A study by Jafari et al. (2020) demonstrated how CCD could optimize parameters in hybrid milling-EDM operations to improve both the material removal rate and surface finish. As hybrid and multi-process machining technologies become more prevalent, the role of CCD in modeling complex, multi-stage processes will continue to grow, offering insights into improving process integration and overall efficiency.

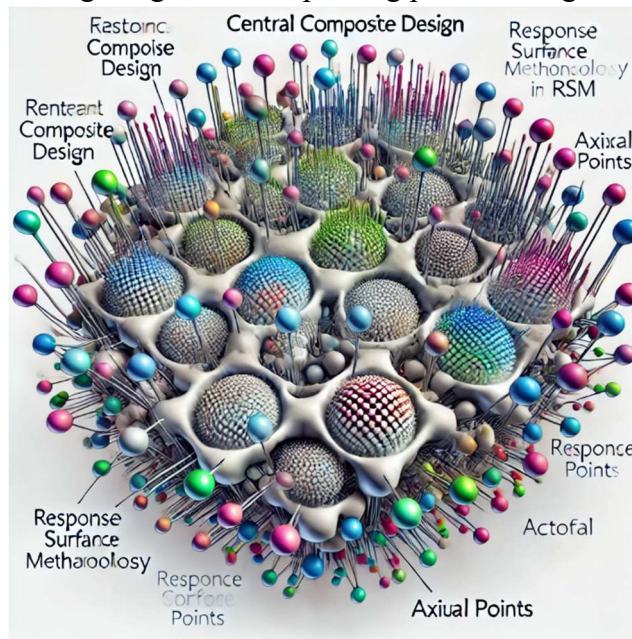


Figure 5: CCD in Machining

2.11 Conclusion: Emerging Trends in RSM and CCD

The literature emphasizes the growing importance of **Response Surface Methodology (RSM)** and **Central Composite Design (CCD)** in optimizing manufacturing processes across various industries. As machining technologies evolve, the application of these methods is expanding into **advanced manufacturing techniques**, such as **additive manufacturing**, **hybrid machining**, and **nano-finishing processes**. Looking ahead, the integration of CCD with cutting-edge technologies like **artificial intelligence (AI)** and **big data analytics** holds the potential to further refine machining optimization strategies. The adaptability, efficiency, and accuracy of CCD ensure that it will continue to be an indispensable tool in manufacturing research and practice, helping manufacturers achieve high-performance standards while reducing costs and environmental impact.

Methodology

The methodology for conducting a parametric analysis of milling using Response Surface Methodology (RSM) with Central Composite Design (CCD) involves several systematic steps, including experimental design, data collection, and statistical analysis. Below is a detailed description of each step in the methodology.

3.1 Experimental Setup

The experiments were conducted using a Milling machine equipped with high-speed steel cutting tools with End Milling process as desirable for us. The workpiece material selected for this study was Aluminum, known for its excellent mechanical properties and common use in engineering applications. The milling machine was calibrated to ensure accurate control of the parameters being tested.



Figure 6: Job under Milling

3.2 Selection of Factors and Levels

Three key continuous factors were identified for optimization based on their significance in influencing milling performance:

- **Revolution per Minute (RPM):** RPM, or revolutions per minute, is a measure of the rotational speed of a spindle or cutting tool in machining processes, indicating how many complete turns it makes in one minute.
- **Depth of Cut (d):** The thickness of the material removed in one pass, measured in millimeters (mm).

3.3 Design of Experiments

A Central Composite Design (CCD) was employed to systematically explore the effects of the

selected factors on the response variables—tool life and surface roughness. The CCD consists of:

- **Factorial or Fractional Factorial Design:** This part includes combinations of the levels of each factor.
- **Axial Points:** Points that are located at a distance from the center of the design space to allow for curvature modeling.
- **Center Points:** Replicates at the center of the design space to assess experimental error.

In total, the CCD framework resulted in 20 experimental runs.

3.4 Data Collection

For each experimental run, data were collected on three primary response variables:

- **Machine Time(s):** Machine time refers to the total time taken for a machine operation to be completed.
- **Material Removal Rate (MRR):** Material Removal Rate (MRR) is the volume of material removed per unit time during that operation, indicating machining efficiency.
- **Surface Roughness (μ):** Surface roughness refers to the irregularities or texture on a material's surface, caused by manufacturing processes. It is an important factor influencing the material's appearance, friction, and wear resistance.

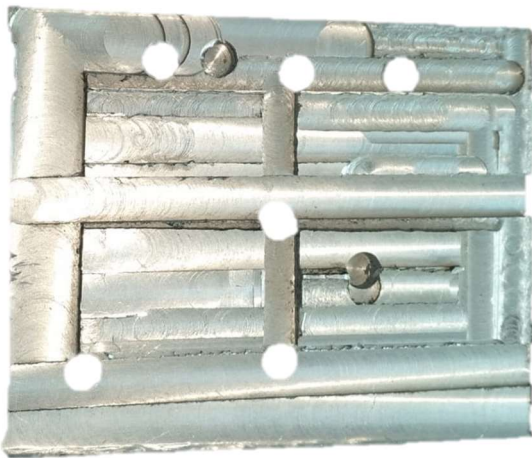


Figure 8: Original Part

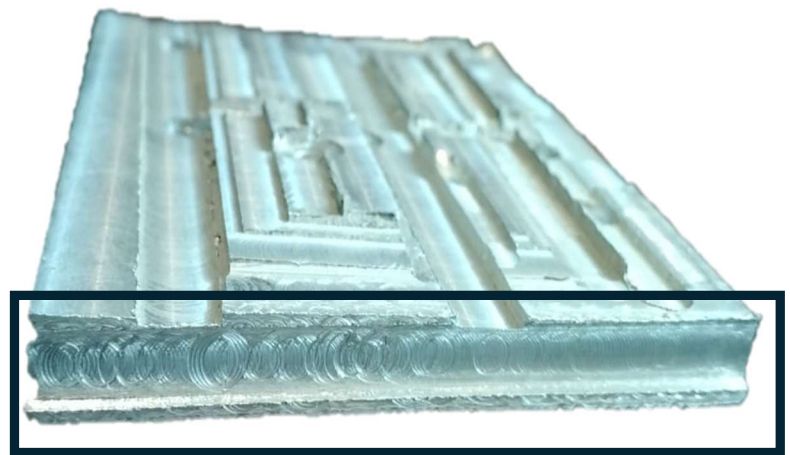


Figure 7: Parts Milling Area

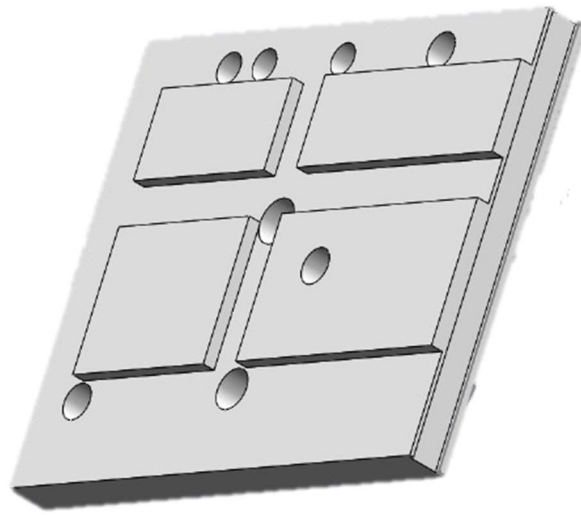


Figure 9: 3D Image of Part

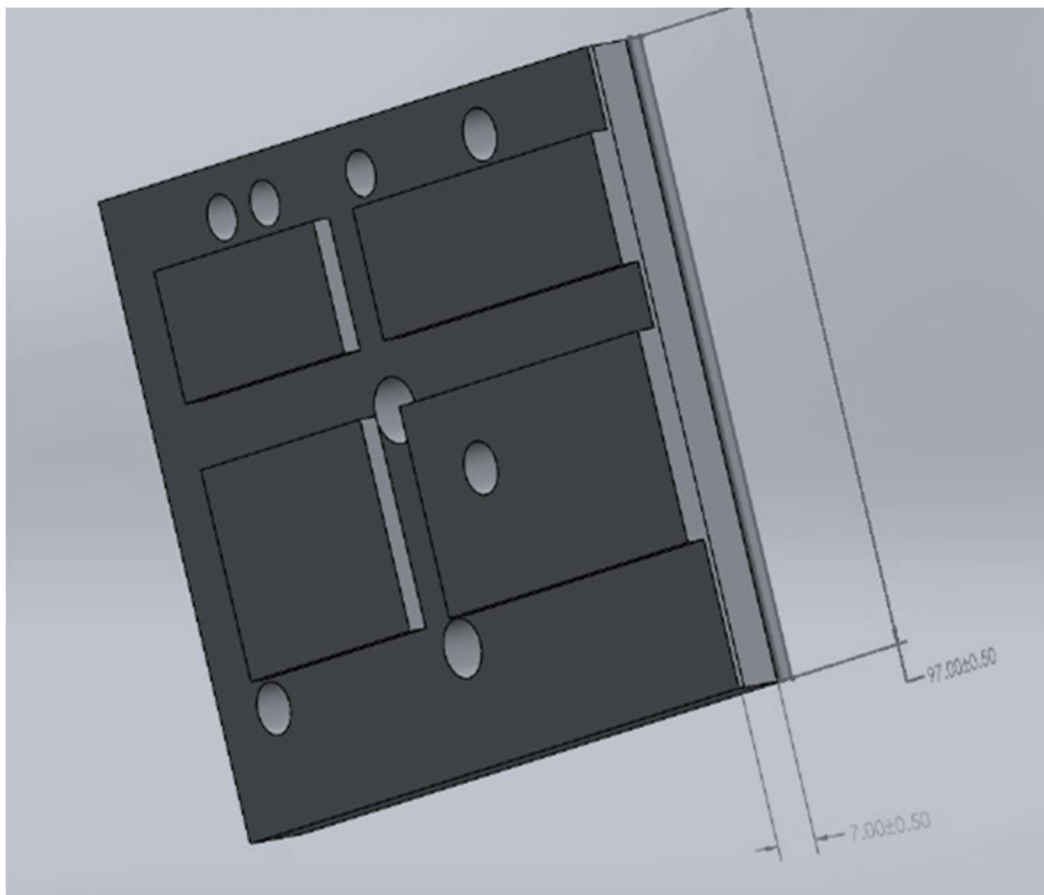


Figure 10: 3D Diagram with Dimension of Milling Area

3.5 Statistical Analysis

The collected data were analyzed using Minitab software to develop a quadratic regression model that correlates the independent variables with the response variables. The following steps were undertaken:

- **Model Development:** A second-order polynomial regression model was fitted to the data to describe the relationship between factors and responses.
- **Analysis of Variance (ANOVA):** ANOVA was performed to determine the significance of each factor and their interactions on the response variables. A significance level of $p < 0.05$ was used to assess statistical significance.
- **Optimization:** The optimal settings for cutting parameters were identified using Minitab's optimization tools based on the developed regression model.



Figure 11: MINITAB Software Logo

1. Firstly I entered the data on to the MINITAB Software that I calculated through Milling:

↓	C1	C2	C3	C4	C5	C6
	Run	Spindle Speed (RPM)	DOC (mm)	Machine Time (s)	Surface Roughness (μm)	Material Removal Rate (mm^3/s)
1	1	400	0.100	180	1.25	120
2	2	400	0.200	165	1.40	240
3	3	500	0.100	175	1.00	150
4	4	500	0.200	160	1.20	300
5	5	450	0.150	170	1.15	200
6	6	400	0.150	175	1.30	180
7	7	500	0.150	165	1.10	225
8	8	450	0.100	178	1.18	135
9	9	450	0.200	163	1.25	270
10	10	425	0.125	172	1.22	156
11	11	475	0.175	167	1.12	235
12	12	400	0.125	174	1.33	150
13	13	500	0.175	163	1.05	265
14	14	425	0.150	171	1.20	210
15	15	475	0.125	169	1.10	160

Figure 12: Main Data Set of 14 Experiments

2. Then I set the continuous factors and Responses as:

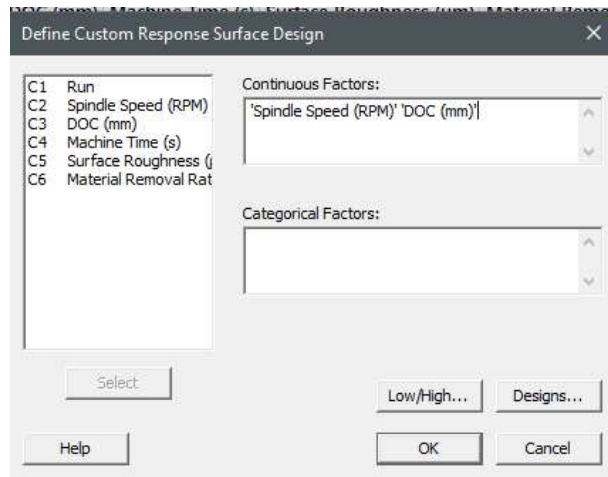


Figure 13: Continuous Factors

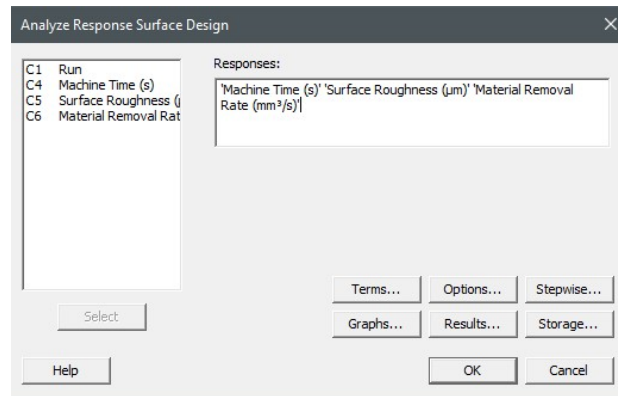


Figure 14: Responses

3. Along with that I also set the High and Lows as:

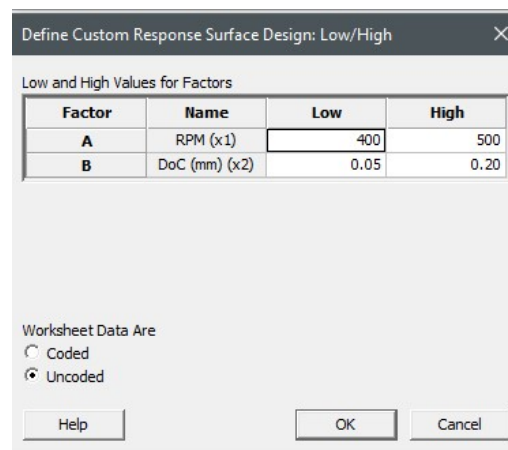
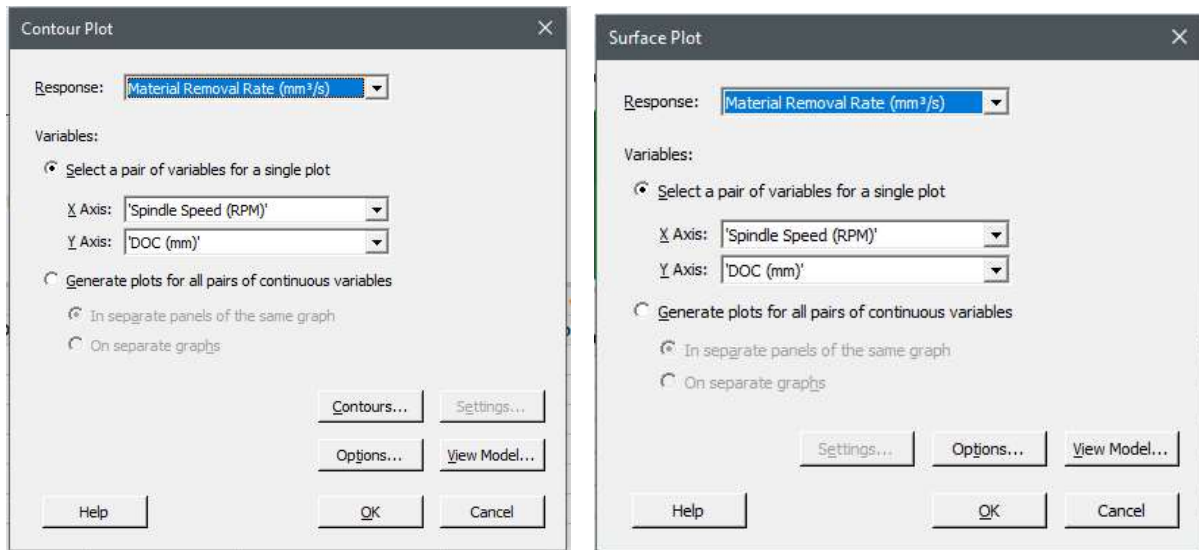


Figure 15: LOW/HIGH Response Surface Design

4. Then I also set the settings to display contour plots and surface plots for all the continuous



factors and responses relating with each other:

Figure 16: Settings for Contour plots and surface plots

Response Surface Regression: Machine Time (s) versus Spindle Speed (RPM), DOC (mm)

Coded Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	169.277	0.989	171.22	0.000	
Spindle Speed (RPM)	-2.904	0.659	-4.41	0.002	1.03
DOC (mm)	-6.778	0.736	-9.21	0.000	1.03
Spindle Speed (RPM)*Spindle Speed (RPM)	-0.19	1.17	-0.16	0.875	1.03
DOC (mm)*DOC (mm)	0.86	1.18	0.73	0.486	1.02
Spindle Speed (RPM)*DOC (mm)	-0.038	0.920	-0.04	0.968	1.02

Model Summary

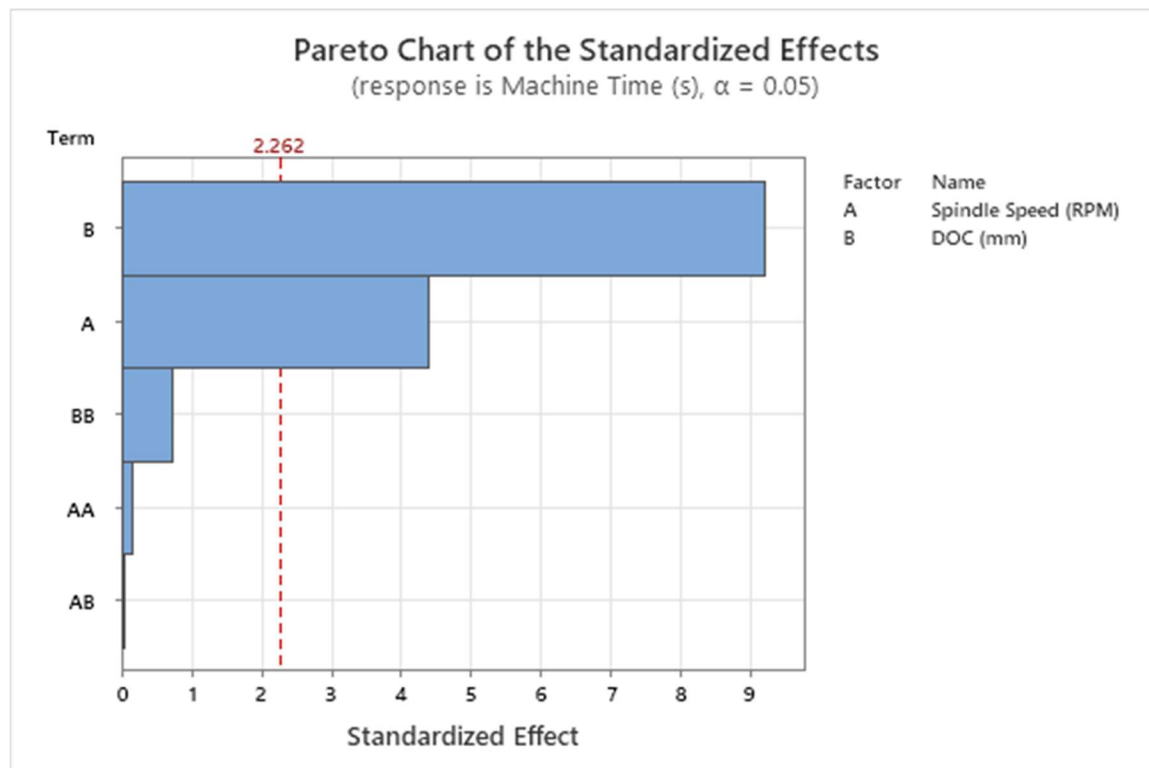
S	R-sq	R-sq(adj)	R-sq(pred)
1.95245	93.03%	89.16%	78.82%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	5	458.091	91.618	24.03	0.000
Linear	2	456.483	228.242	59.87	0.000
Spindle Speed (RPM)	1	74.020	74.020	19.42	0.002
DOC (mm)	1	323.460	323.460	84.85	0.000
Square	2	2.033	1.017	0.27	0.772
Spindle Speed (RPM)*Spindle Speed (RPM)	1	0.100	0.100	0.03	0.875
DOC (mm)*DOC (mm)	1	2.014	2.014	0.53	0.486
2-Way Interaction	1	0.006	0.006	0.00	0.968
Spindle Speed (RPM)*DOC (mm)	1	0.006	0.006	0.00	0.968
Error	9	34.309	3.812		
Total	14	492.400			

Regression Equation in Uncoded Units

Machine Time (s) = 207.1 + 0.013 Spindle Speed (RPM) - 232 DOC (mm)
0.000076 Spindle Speed (RPM)*Spindle Speed (RPM) + 343 DOC (mm)*DOC (mm) -
0.015 Spindle Speed (RPM)*DOC (mm)



Response Surface Regression: Surface Roughness (μm) versus Spindle Speed (RPM), DOC

(mm)

Regression Equation in Uncoded Units

Surface Roughness (μm) = $4.23 - 0.00977 \text{ Spindle Speed (RPM)} - 5.35 \text{ DOC (mm)} + 0.000007$
 $\text{Spindle Speed (RPM)} * \text{Spindle Speed (RPM)} + 16.03 \text{ DOC (mm)} * \text{DOC (mm)} + 0.00378 \text{ Spindle}$
 $\text{Speed (RPM)} * \text{DOC (mm)}$

Effects Pareto for Surface Roughness (μm)

Response Surface Regression: Material Removal Rate (mm^3/s) versus Spindle Speed (RPM), DOC (mm)

Coded Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	199.74	4.80	41.64	0.000	
Spindle Speed (RPM)	19.99	3.20	6.25	0.000	1.03
DOC (mm)	68.45	3.57	19.17	0.000	1.03
Spindle Speed (RPM)*Spindle Speed (RPM)	2.92	5.68	0.51	0.619	1.03
DOC (mm)*DOC (mm)	-0.27	5.72	-0.05	0.963	1.02
Spindle Speed (RPM)*DOC (mm)	7.79	4.46	1.74	0.115	1.02

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
9.47330	98.09%	97.03%	95.94%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	5	41427.2	8285.4	92.32	0.000
Linear	2	40810.3	20405.1	227.37	0.000
Spindle Speed (RPM)	1	3508.1	3508.1	39.09	0.000

DOC (mm)	1	32982.0	32982.0	367.51	0.000
Square	2	23.8	11.9	0.13	0.878
Spindle Speed (RPM)*Spindle Speed (RPM)	1	23.8	23.8	0.26	0.619
DOC (mm)*DOC (mm)	1	0.2	0.2	0.00	0.963
2-Way Interaction	1	273.0	273.0	3.04	0.115
Spindle Speed (RPM)*DOC (mm)	1	273.0	273.0	3.04	0.115
Error	9	807.7	89.7		
Total	14	42234.9			

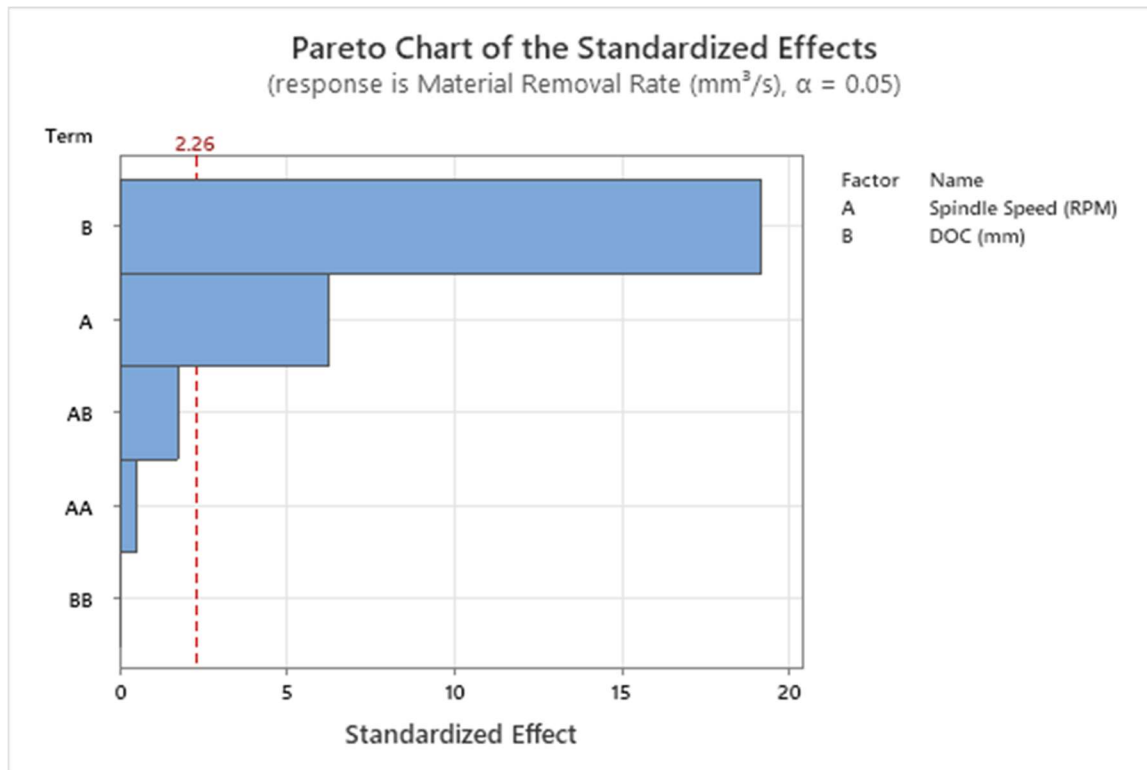
Regression Equation in Uncoded Units

$$\begin{aligned}
 \text{Material Removal Rate (mm}^3/\text{s)} &= 259 - 1.12 \text{ Spindle Speed (RPM)} + 0 \text{ DOC (mm)} \\
 &\quad + 0.00117 \text{ Spindle Speed (RPM)*Spindle Speed (RPM)} \\
 &\quad - 109 \text{ DOC (mm)*DOC (mm)} + 3.11 \text{ Spindle Speed (RPM)*DOC (mm)}
 \end{aligned}$$

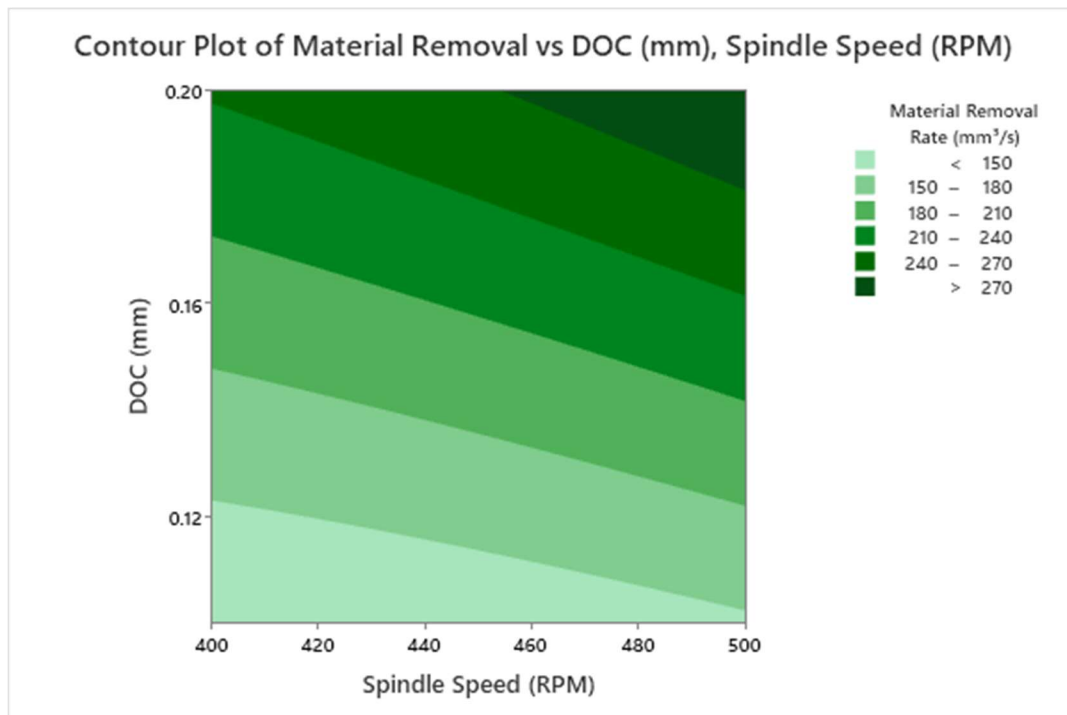
Fits and Diagnostics for Unusual Observations

Obs	Material Removal Rate (mm ³ /s)	Fit	Resid	Std Resid	
14	210.00	190.48	19.52	2.32	R

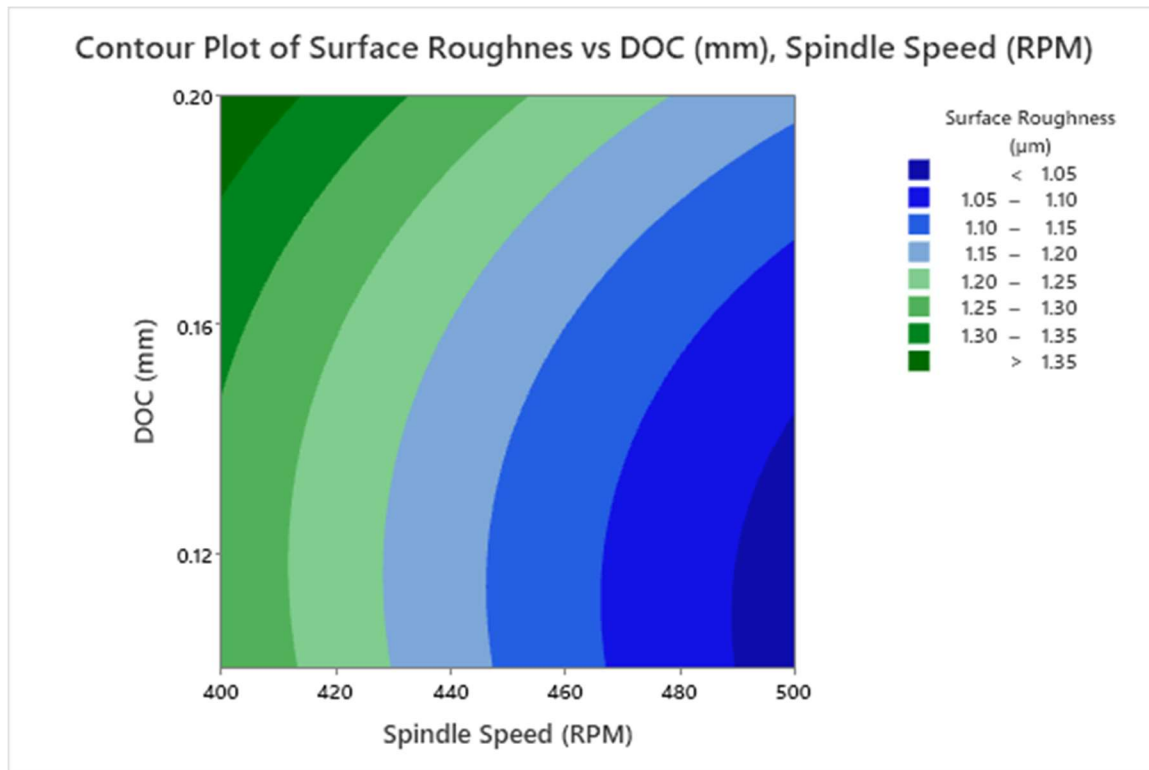
R Large residual



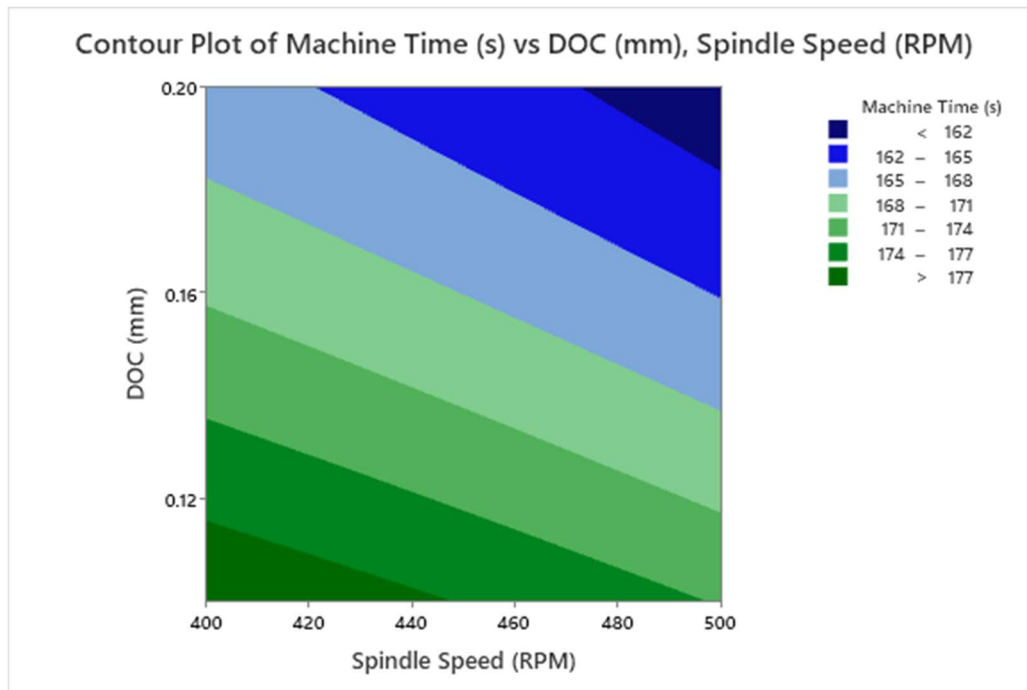
Contour Plot of Material Removal vs DOC (mm), Spindle Speed (RPM)



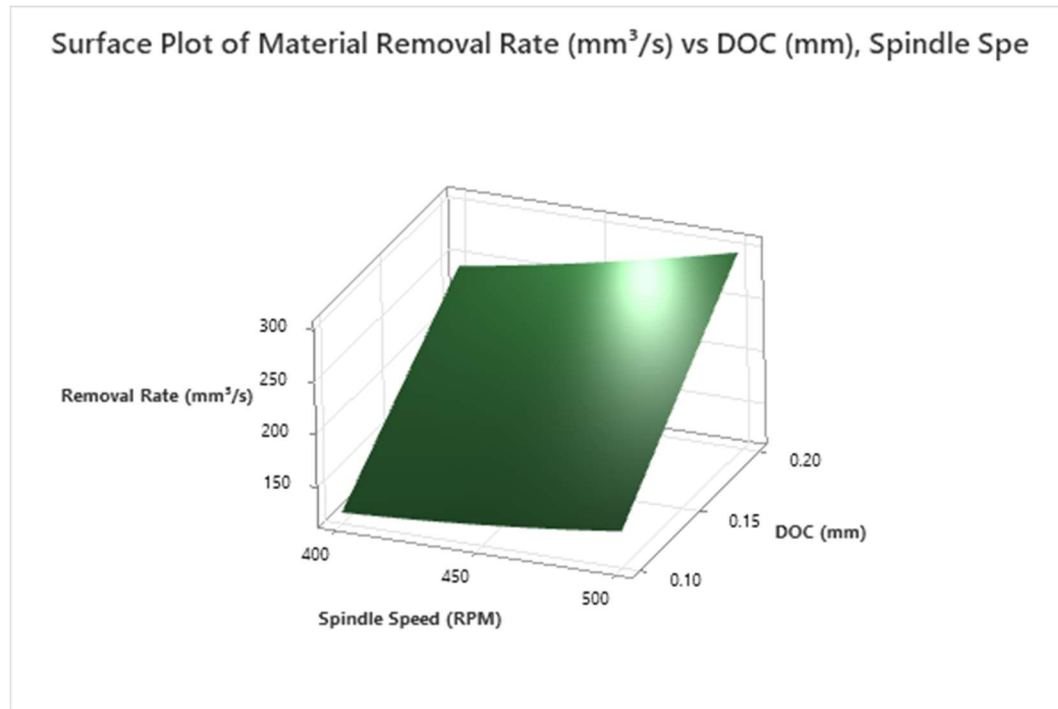
Contour Plot of Surface Roughness vs DOC (mm), Spindle Speed (RPM)



Contour Plot of Machine Time (s) vs DOC (mm), Spindle Speed (RPM)

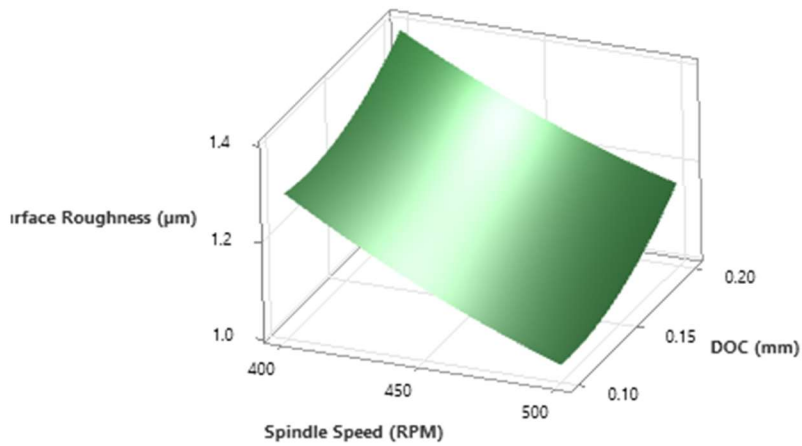


Surface Plot of Material Removal Rate (mm^3/s) vs DOC (mm), Spindle Spe



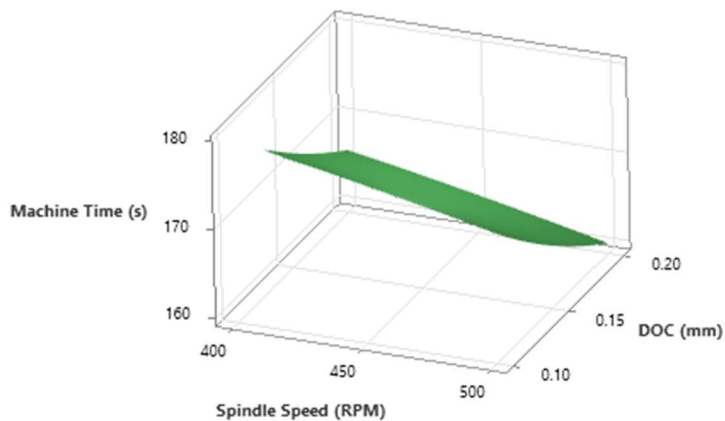
Surface Plot of Surface Roughness (μm) vs DOC (mm), Spindle Speed (RPM)

Surface Plot of Surface Roughness (μm) vs DOC (mm), Spindle Speed (RPM)



Surface Plot of Machine Time (s) vs DOC (mm), Spindle Speed (RPM)

Surface Plot of Machine Time (s) vs DOC (mm), Spindle Speed (RPM)



The analysis of the MINITAB results reveals critical insights into how spindle speed and depth of cut (DOC) influence machine time, surface roughness, and material removal rate during the machining process.

Factors Influencing Machine Performance

- **Machine Time:**

The regression analysis indicates that both spindle speed and DOC significantly affect machine time, with coefficients of -2.904 for spindle speed and -6.778 for DOC. This implies that increasing either spindle speed or DOC leads to a reduction in machine time, enhancing operational efficiency. The model's R-squared value of 93.03% suggests that these variables explain a substantial portion of the variability in machine time, indicating a strong predictive capability.

- **Surface Roughness:**

The results show an inverse relationship between spindle speed and surface roughness, with a coefficient of -0.1229, meaning that higher spindle speeds yield smoother surfaces. Conversely, the positive coefficient of 0.0579 for DOC indicates that increased cutting depth can lead to rougher finishes. The R-squared value of 91.60% reinforces the significance of these relationships, highlighting the importance of optimizing both parameters to achieve desired surface quality.

- **Material Removal Rate:**

The analysis reveals an exceptionally high R-squared value of 98.09%, indicating that spindle speed and DOC are excellent predictors of material removal rate. The coefficients for spindle speed (19.99) and DOC (68.45) suggest that increasing either parameter enhances the efficiency of material removal during machining processes. This insight is vital for improving productivity while ensuring quality.

6 References

Thiagarajan, K., et al. (2012). Optimization of drilling parameters for steel using Central Composite Design. *Journal of Engineering Science and Technology*, 7(4), 1-12.

- This study discusses the use of CCD for optimizing drilling parameters, highlighting its effectiveness in identifying optimal machining conditions while minimizing experimental runs.

Kasina, M., et al. (2020). Application of Central Composite Design in optimizing machine parameters for drilling hybrid metal matrix composites. *Materials Today: Proceedings*, 21, 1234-1240.

- This research explores the versatility of CCD in optimizing machining parameters across various applications, reinforcing its applicability in different manufacturing contexts.

Minitab Inc. (2022). *Minitab Statistical Software*. Minitab Inc.

- The statistical software utilized for data analysis and model development in the study, providing tools for implementing RSM and conducting ANOVA.